

WESTERN HEAD, NS, Canada

WMO: 714110

Lat: 43.990N

Lon: 64.664W

Elev: 10

StdP: 101.20

Time Zone: -4.00 (NAA)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.9	-12.7	-20.6	0.6	-13.2	-18.3	0.7	-11.0	15.9	5.2	14.0	4.4	4.1	300	0.657

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.5	25.0	18.1	22.5	17.5	20.7	17.3	20.1	22.2	19.0	20.9	18.2	19.8	4.8	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.4	14.1	21.0	18.4	13.2	19.7	17.5	12.6	18.9	57.7	22.2	54.0	21.0	51.4	19.8	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.5	10.7	9.3	-18.1	30.6	2.9	2.4	-20.2	32.3	-21.8	33.7	-23.4	35.1	-25.5	36.8		
(5)			-18.2	23.3	2.7	1.7	-20.2	24.5	-21.8	25.5	-23.3	26.4	-25.3	27.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.1	-2.9	-2.6	0.3	4.6	8.7	12.9	15.7	16.5	14.9	10.4	5.4	0.6
	DBStd	7.72	5.20	4.54	3.87	2.81	2.80	2.91	2.61	2.30	2.79	3.47	4.06	4.64
	HDD10.0	1759	400	353	302	162	61	5	0	0	1	38	144	293
	HDD18.3	4133	658	586	559	411	299	167	91	67	108	247	388	551
	CDD10.0	692	0	0	1	1	19	91	176	200	148	49	6	0
	CDD18.3	24	0	0	0	0	0	2	8	9	5	1	0	0
	CDH23.3	162	0	0	1	0	7	32	68	39	14	2	0	0
	CDH26.7	37	0	0	0	0	1	8	19	7	2	0	0	0
Wind	WSAvg	4.3	4.9	4.8	4.8	4.5	3.8	3.6	3.2	3.4	3.9	4.6	4.7	4.9
Precipitation	PrecAvg	1405	124	132	123	110	111	84	74	108	100	120	159	161
	PrecMax	1951	220	233	194	174	198	148	192	219	326	269	364	246
	PrecMin	873	39	65	33	28	34	43	21	30	22	47	61	91
	PrecStd	274	40	51	44	39	50	28	46	62	75	60	77	51
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.4	8.5	13.4	17.2	23.1	27.1	29.2	27.2	25.1	21.2	15.5	11.8
		MCWB	8.1	6.3	7.4	9.5	14.2	17.7	19.9	19.6	19.2	15.7	12.9	10.7
	2%	DB	7.5	6.2	8.7	13.2	17.9	22.8	24.8	23.9	22.3	18.3	13.7	10.2
		MCWB	6.4	5.0	5.0	7.5	11.9	15.4	18.2	18.5	18.5	15.6	12.2	9.4
	5%	DB	6.1	4.7	6.5	10.4	14.8	19.4	21.5	21.6	20.6	16.9	12.5	8.8
		MCWB	5.1	3.6	4.4	6.8	10.5	14.6	18.0	17.9	17.5	15.1	11.1	8.1
	10%	DB	4.6	3.5	5.1	8.6	12.5	16.8	19.4	20.2	19.2	15.8	11.5	7.4
		MCWB	3.5	2.4	3.5	6.2	9.7	14.1	17.3	17.8	16.8	14.1	10.2	6.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.7	7.1	8.8	11.4	16.4	19.1	22.3	22.0	21.1	18.0	14.2	11.3
		MCDB	8.9	8.1	11.0	13.9	20.2	24.8	26.6	25.4	23.0	19.2	14.5	11.5
	2%	WB	6.8	5.3	6.2	9.0	12.9	16.8	19.6	20.0	19.6	16.7	12.8	9.6
		MCDB	7.2	5.9	7.6	11.1	16.2	20.3	22.2	21.8	21.0	17.6	13.2	9.9
	5%	WB	5.3	3.8	4.9	7.8	11.2	15.5	18.4	18.9	18.6	15.6	11.7	8.3
		MCDB	5.8	4.5	6.0	9.6	13.5	18.2	20.5	20.6	19.9	16.5	12.2	8.7
	10%	WB	3.7	2.5	3.7	6.7	10.1	14.4	17.5	18.1	17.5	14.5	10.5	6.7
		MCDB	4.4	3.3	4.8	8.1	11.9	16.4	19.3	19.7	18.7	15.5	11.2	7.3
Mean Daily Temperature Range	5% DB	MDBR	7.9	7.7	7.3	7.2	7.1	7.3	7.4	7.5	8.0	8.1	7.2	7.4
		MCDBR	8.7	7.9	8.9	11.1	12.3	13.2	12.1	10.4	9.7	8.9	7.6	8.0
		MCWBR	8.7	7.2	6.8	7.2	7.4	7.3	7.1	6.4	6.4	6.6	6.9	7.9
	5% WB	MCDBR	8.6	7.6	8.1	8.8	9.9	10.7	9.6	8.3	7.6	7.3	7.0	7.6
		MCWBR	8.9	7.5	7.0	6.7	6.8	6.8	6.8	5.9	5.7	6.3	7.0	8.0
Clear-Sky Solar Irradiance	taub	0.306	0.304	0.332	0.359	0.380	0.406	0.423	0.402	0.363	0.341	0.323	0.308	
	taud	2.487	2.513	2.456	2.423	2.406	2.364	2.337	2.390	2.461	2.528	2.534	2.509	
	Ebn at Noon	825	897	908	906	891	866	846	853	863	840	799	786	
	Edh at Noon	70	81	98	109	115	120	122	112	97	80	67	63	
All-Sky Solar Radiation	RadAvg	1.40	2.28	3.54	4.40	5.04	5.56	5.46	5.05	4.18	2.70	1.54	1.14	
	RadStd	0.10	0.19	0.28	0.40	0.61	0.53	0.38	0.34	0.24	0.29	0.17	0.10	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	+0.93	N/A	N/A	N/A	+104	+12
(47) Regional (28 neighbors)		N/A	N/A	N/A	N/A	+0.42	+0.55	N/A	+58	+7

Nomenclature: See separate page